

Fast recovery, high sensitivity NMR probe and preamplifier for low frequencies

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Fast recovery, high sensitivity NMR probe and preamplifier for low frequencies

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The design of a probe-preamplifier combination which reduces the NMR receiver recovery time following a pulse by a factor of at least twenty at 5 MHz is described. The preamplifier, which has a noise figure of 1.3 dB, employs negative feedback to damp the probe coil (a typical effective Q is 7), but maintains the signal-to-noise ratio that one would expect from an undamped coil. In addition, the use of a short phase-inverted pulse following the main transmitter pulse serves to reduce drastically the ring-down of the probe for those high voltages where the preamplifier is inoperative. Optimum utilization of transmitter power in achieving fast pulse rise times is also discussed, as well as elimination of "phase-glitch" by proper tuning and the use of a broadband transmitter.

INTRODUCTION

It has long been recognized that operation of a pulsed nuclear magnetic resonance (NMR) spectrometer at low frequencies presents special problems if the signal-to-noise ratio of the experiment is not high, and a large bandwidth is also required. Briefly stated, the difficulty encountered is that while sensitivity is proportional to the square root of the inherent quality factor (Q) of the receiving coil, the receiver recovery time is also dependent on Q , and in an attempt to obtain high sensitivity by increasing Q , a fast decaying transient signal may be obliterated by the ringing of the probe circuit subsequent to the transmitter pulse. The dilemma is illustrated in Fig. 1, and typically one must wait at least fifteen times the time constant of the ringing before the signal may be observed without distortion. Several schemes have been proposed to reduce this "ring-down" time, with varying degrees of success.¹⁻⁷ Many involve temporarily damping the probe circuit to accelerate ring-down followed by removal of the damping to restore the circuit's Q and hence sensitivity. Two problems arise with this approach. First, the act of removing the damping causes the circuit to ring anew, for the change in voltage required (to operate the damping switch in a time short in comparison with that of interest) inevitably contains Fourier components which excite the probe. These components are usually transmitted to the probe by stray capacitance, and even if the damping is removed relatively slowly, an excitation of several millivolts still results which requires a period of, say, eight time constants to die away into the noise. Second, if a switch with perfect isolation could be made, distortion of the initial part of the free induction decay (FID) would then become apparent. The finite bandwidth of the probe imposes a limited rise-time upon the signal. If the time constant T_2 of the FID is not long in comparison with that of the probe, τ , then measurements

of T_2 are in error and upon Fourier transformation, distortion of the line shape results. This is shown in Fig. 2.

It follows that in order to reduce substantially the ring-down time and eliminate distortion, one has no option but to damp the probe permanently, and we therefore require a method of damping which does not introduce extra noise. As damping must by its very nature involve energy dissipation, i.e., resistance, the only way to avoid Johnson noise is to reduce the temperature of the damping resistance. This may be accomplished in two ways. An obvious and somewhat impractical method is to place the damping resistance in liquid helium. A decrease in Q factor of nearly 20 may then be obtained for a loss of only 1 dB in noise figure. However, a more subtle method employs negative feedback to produce an amplifier which has a low input impedance and also a low noise figure; the low impedance then damps the tuned circuit. We therefore assume that in some way (to be discussed later) the ringing of the probe is damped down to a few millivolts, and that from this point on, the amplifier, with its low input impedance, takes over the damping function.

I. NEGATIVE FEEDBACK NOISE PERFORMANCE

Consider the operation of the circuit of Fig. 3. X represents the impedance of the probe with its attendant noise N , while Y represents the impedance of the feedback path with its attendant noise E_f . The amplifier has a gain of $+\alpha$, which may be complex, and an output V , and the noise voltage and current sources associated with the amplifier are given by E_a and I_a , respectively. E_a and I_a are assumed to be uncorrelated. (A full description of this method of amplifier noise analysis may be found in Ref. 8.) Examining the potential around the loop ABCD at any instant of time, we find that

$$V/\alpha - E_a + E_f - V = (I_a + I)Y, \quad (1)$$

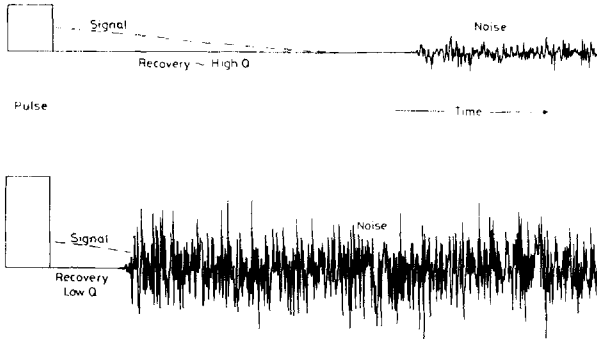


FIG. 1. Reducing the quality factor of the receiving coil reduces the receiver recovery time, enabling a signal to be viewed, but it also reduces the signal-to-noise ratio.

while between F and A ,

$$N - IX + E_a = V/\alpha. \quad (2)$$

Hence

$$\begin{aligned} -V \left[1 - \frac{1}{\alpha} \left(1 + \frac{Y}{X} \right) \right] \\ = I_a Y + \frac{Y}{X} N + E_a \left(1 + \frac{Y}{X} \right) - E_f. \end{aligned} \quad (3)$$

Thus, in general, the mean square noise voltage out of the amplifier is given by

$$\begin{aligned} \overline{V^2} = \{ I_a^2 Y Y^* + (Y Y^* / X X^*) N^2 + E_a^2 (1 + Y/X)(1 + Y^*/X^*) \\ + E_f^2 \} \left[1 - \frac{1}{\alpha} \left(1 + \frac{Y}{X} \right) \right] \left[1 - \frac{1}{\alpha^*} \left(1 + \frac{Y^*}{X^*} \right) \right]^{-1}, \end{aligned} \quad (4)$$

where the use of an asterisk denotes a complex conjugate and the noise voltages are now rms values. If the amplifier were noiseless, (i.e., $I_a = E_a = E_f = 0$) the output would be

$$\begin{aligned} \overline{V^2} = \frac{Y Y^*}{X X^*} N^2 \left[1 - \frac{1}{\alpha} \left(1 + \frac{Y}{X} \right) \right] \\ \times \left[1 - \frac{1}{\alpha^*} \left(1 + \frac{Y^*}{X^*} \right) \right]^{-1}. \end{aligned} \quad (5)$$

Thus the ratio of the actual noise output to the perfect noise output, which is the noise factor, is given by, from Eqs. (4) and (5),

$$\begin{aligned} F = \left[I_a^2 X X^* + N^2 + E_a^2 \left(1 + \frac{X}{Y} \right) \right. \\ \left. + \frac{X^*}{Y^*} + \frac{X X^*}{Y Y^*} \right] / N^2. \end{aligned} \quad (6)$$

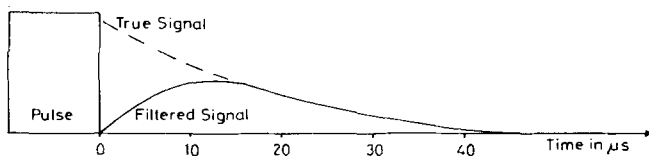


FIG. 2. Even if instantaneous recovery of a high Q probe were possible, the signal would still be distorted due to the finite bandwidth of the probe.

Note that if the amplifier gain be sufficiently large that E_a and I_a are independent of α , the noise factor is also independent of α . Let us consider first the effects of a capacitance C' in the feedback loop. Thus $Y = (j\omega C')^{-1}$ and $E_f = 0$, $j = \sqrt{-1}$. The probe impedance X may be expressed in the Lorentzian form

$$X = Q\omega_0 L / (1 + j\Delta\omega\tau), \quad \tau = 2Q/\omega_0, \quad (7)$$

where Q is the quality factor of the probe, ω_0 is the resonant frequency (presumably the Larmor frequency of interest), L is the coil inductance, $\Delta\omega$ is the frequency distance off resonance, and τ is the time constant of the probe ringing. The noise associated with the real part of the probe impedance is given, for a temperature T and bandwidth $\Delta f \ll \omega_0/Q$, by

$$N^2 = 4kT\Delta f Q\omega_0 L / (1 + \Delta\omega^2\tau^2). \quad (8)$$

Substituting in Eq. (6) to find the noise factor we obtain, after some manipulation,

$$\begin{aligned} F = \{ I_a^2 (Q\omega_0 L)^2 + 4kT\Delta f (Q\omega_0 L) \\ + E_a^2 [1 + \omega_0^4 C'^2 L^2 Q^2 + \Delta\omega (4\omega_0 C' L Q^2 + 2\omega_0^3 C'^2 L^2 Q^2) \\ + \Delta\omega^2 (4Q^2/\omega_0^2 + 4Q^2 L C' + \omega_0^2 L^2 C'^2 Q^2)] \} \\ \times [4kT\Delta f (Q\omega_0 L)]^{-1}, \end{aligned} \quad (9)$$

and this is a minimum, given by

$$\begin{aligned} F = [I_a^2 (Q\omega_0 L)^2 \\ + 4kT\Delta f (Q\omega_0 L) + E_a^2] / 4kT\Delta f (Q\omega_0 L), \end{aligned} \quad (10)$$

when

$$\Delta\omega = - \left\{ \frac{2\omega_0^3 C' L Q^2 + \omega_0^5 C'^2 L^2 Q^2}{4Q^2 + 4Q^2 \omega_0^2 L C' + \omega_0^4 L^2 C'^2 Q^2} \right\}.$$

For most practical purposes, this reduces to

$$\Delta\omega = -\omega_0^3 L C' / 2, \quad (11)$$

which, in the present situation, is less than the probe bandwidth. To obtain an estimate of the bandwidth over which a good noise factor is maintained, we note from Eq. (9) that the coefficient of E_a^2 is the only one to vary with frequency, and that it doubles when the frequency is, to a high degree of accuracy,

$$\Delta\omega = -\omega_0^3 L C' / 2 \pm \omega_0 / 2Q. \quad (12)$$

The preferred bandwidth is therefore of the order of

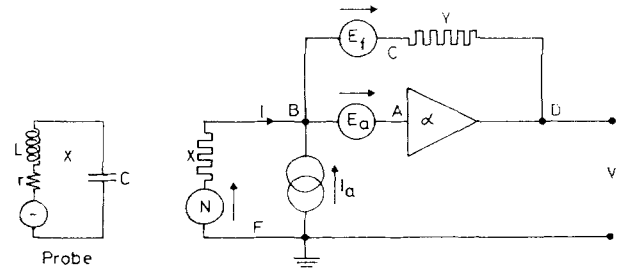


FIG. 3. The noise equivalent circuit of an amplifier with negative feedback. The amplifier has a gain α and the noise associated with it is represented by E_a and I_a . The noise E_f is associated with the real part of the feedback impedance Y , while the noise N is associated with the real part of the probe impedance X .

ω_0/Q . This is the same value that would be obtained if no feedback were applied, and it is worth noting that the minimum noise factor of Eq. (10) is also the same as if no feedback were applied. We therefore conclude that the only effect of capacitive feedback upon the noise performance is to change slightly the frequency at which the best sensitivity is obtained. It neither degrades the noise performance nor changes the preferred bandwidth.

Let us now repeat the above calculations with resistive feedback. Thus $Y = R$ and $E_f^2 = 4kT R \Delta f$. Substituting in Eq. (6) we find that

$$F = \{I_a^2(Q\omega_0 L)^2 + 4kT \Delta f(Q\omega_0 L)(1 + Q\omega_0 L/R) + E_a^2[(1 + Q\omega_0 L/R)^2 + \Delta\omega^2 \tau^2]\} \times [4kT \Delta f(Q\omega_0 L)]^{-1}. \quad (13)$$

It is immediately apparent that the best noise performance is obtained when $\Delta\omega = 0$ and $R \gg 2Q\omega_0 L$. If these conditions hold, then the preferred bandwidth is once again of the order of ω_0/Q . It is clear, from a comparison of Eqs. (10) and (13) that capacitive feedback is capable of giving a slightly better noise performance than resistive feedback, and further, it will be seen that the absence of any constraint such as $R \gg 2Q\omega_0 L$ simplifies the design procedure of a suitable amplifier. Unfortunately however, neither feedback method, in contrast to a low-temperature resistance, increases the preferred bandwidth.

II. FEEDBACK DAMPING

If the open-loop input impedance of an amplifier is very high, then the closed-loop input impedance, assuming as before a feedback impedance of Y and a gain of α , is given by

$$Z_{in} = Y/(1 - \alpha). \quad (14)$$

Consider first resistive feedback ($Y = R$). If the probe is to be damped by say an order of magnitude, the input impedance must be resistive and have a value of $Q\omega_0 L/10$. $Q\omega_0 L$ is of course the parallel resistance of the probe. Now we have already seen that for good noise performance, $R \gg 2Q\omega_0 L$. Thus, from Eq. (14),

$$Q\omega_0 L(1 - \alpha)/10 = R \gg 2Q\omega_0 L. \quad (15)$$

Thus $(-\alpha)$ must be very much greater than 19 and must be real. At this point, it is instructive to consider a few practical values. Typically, $Q\omega_0 L \sim 10 \text{ k}\Omega$. Thus R is at least $100 \text{ k}\Omega$ and α is at least -100 . Now one cannot avoid the effects of stray capacitance; there will, for example, be some in the feedback loop associated with R , say 1 pF or less. At 10 MHz 1 pF has a reactance of about $15 \text{ k}\Omega$ and cannot be neglected. Worse still, the Miller effect [Eq. (14)] ensures that at the amplifier input, the effective capacitance is 100 pF , and this may well be sufficient to detune the combined preamplifier and probe by say 30% . The positions of maximum signal and minimum noise do not therefore coincide and while the signal-to-noise ratio stays constant about the resonant frequency

of the (free-standing) probe, the amplitudes of both signal and noise vary considerably. Resistive feedback would therefore appear to be a poor method of probe damping, being difficult to implement without great care in amplifier design. Reduction of amplifier gain, leading to an easing of the design problem leads also to a deterioration in the noise performance.

We therefore turn to capacitive feedback, once again requiring that the input impedance of the amplifier be $Q\omega_0 L/10$. If $Y = (j\omega C')^{-1}$, then from Eq. (14),

$$Q\omega_0 L/10 = 1/(1 - \alpha)j\omega C'.$$

Thus

$$\alpha = 1 + 10j/(Q\omega_0 L)\omega C', \quad (16)$$

or, if C is the tuning capacitor for the probe, we have on resonance that

$$\alpha = 1 + (10j/Q)C/C'. \quad (17)$$

For small values of C' , α is almost purely imaginary corresponding to a phase shift of about 90° in the open loop amplifier. For example, for $C = 100 \text{ pF}$, $C' = 1 \text{ pF}$, $Q = 50$, $|\alpha| = 20$, $\arg\alpha = +87^\circ$. Such an amplifier is very easy to build. The starting point is a three stage amplifier with a low frequency open loop gain of $-5.8|\alpha|$. At the frequency of interest, phase shifts of -45° are built into the first and third stages giving the required total gain and phase shift (see Appendix). The phase is correct when the circuit does not detune the probe; in other words, the positions of maximum signal and minimum noise outputs virtually coincide and a change of C' does not alter the resonant frequency. The advantage of an amplifier design giving a phase shift of -270° rather than $+90^\circ$ is that low-frequency stabilization is simple, and at high frequency, one is working with, rather than against, the natural inherent phase shifts caused by device capacitance. This ensures that the design is stable, easy to build and applicable over a large range of frequencies.

III. AMPLIFIER FIRST STAGE

Equation (10) tells us that the minimum noise factor is given by, for a particular value of $Q\omega_0 L$,

$$F = I_a^2(Q\omega_0 L)/4kT \Delta f + E_a^2/4kT \Delta f(Q\omega_0 L) + 1. \quad (18)$$

This is further minimized, giving a value of F of F_{opt} , when

$$Q\omega_0 L = E_a/I_a = R_0. \quad (19)$$

R_0 is the optimum source impedance. Now typically $Q\omega_0 L$ is in the range 2.5 to $25 \text{ k}\Omega$, depending, at any frequency, upon the number of turns in the probe coil. The first stage of amplification therefore requires a device whose optimum source impedance is in this rather high range for the frequency span of interest (1 to 20 MHz). A field effect transistor (FET) is obviously indicated, but the low transconductance of silicon devices in comparison with their bipolar counterparts gives rise to

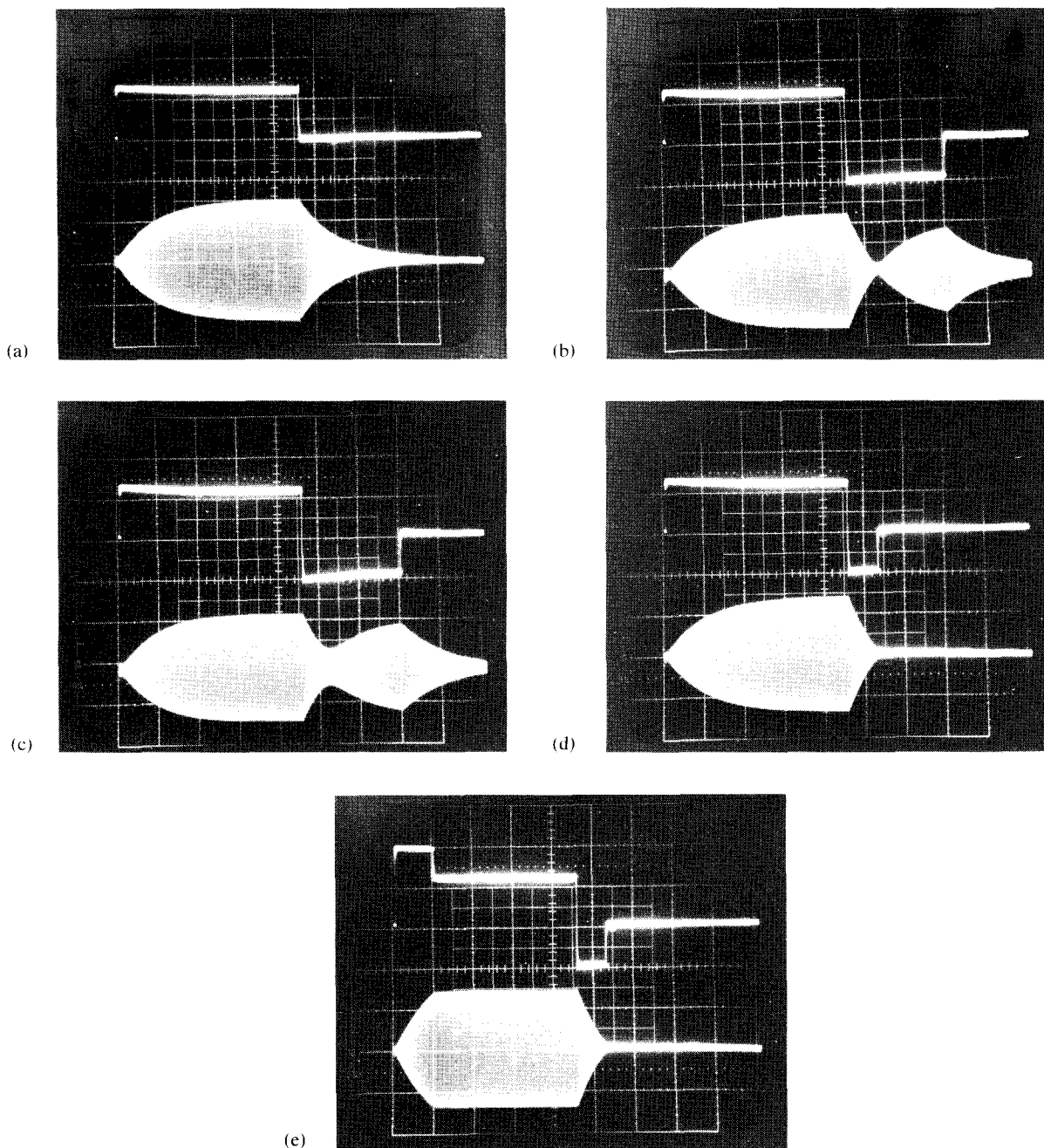


FIG. 8. The effects of various transmitter pulses upon the probe pulse rise and fall times. The upper trace shows the voltage applied to the transmitter modulator and the lower trace the voltage across the probe tuned circuit. (a) shows the long rise and fall times associated with the quality factor of the probe by the relationship $\tau = 2Q_{eff}/\omega_0$. If the transmitter pulse is inverted and the probe is exactly on resonance, the ringing pattern passes through a null, as shown in (b). [A slight detuning of the probe, however, destroys the null (c).] The ring-down may therefore be considerably speeded up (d) by terminating the inverted pulse at the null. If extra transmitter power is available, the rise-time may also be improved in the same way (e). However, a more efficient way of accomplishing this is discussed in the text.

level from a 2 V start. The inductors and the capacitor leading from the probe to the amplifier input resonate at the frequency of interest, and of course the 220 Ω resistor in the attenuator is too small to influence the noise performance.

We must next consider how to speed the ring-down from say 1 kV to 2 V. This would normally take about 6 time constants, or, for the probe employed with Fig. 7, about 30 μ s. Figure 8 shows how this time may be reduced to 0.69 time constants by following the transmitter pulse with a phase-inverted pulse which rapidly cancels

the ringing. (If there is spare transmitter power available, the technique may also be used to speed the pulse rise-time, as shown.) It must be emphasized that employment of this technique places considerable demands on the stability of the equipment. Figure 8(c) shows the effect of moving slightly off-resonance—a null cannot be obtained. It is also very difficult to obtain a null if the transmitter is tuned, for the nulling depends essentially on the absence of “phase-glitch,”¹⁰ and the more tuned stages are employed, the more difficult it becomes exactly to align them all. Figure 8 was obtained with a

broadband transmitting system, and it may be seen from Figs. 8 and 9 that it is possible to avoid "phase-glitch" with such a system. Not only phase stability, but also timing stability is essential for success, and when using very short pulses it may be necessary to employ phase synchronization to ensure reproducibility from pulse to pulse. However, with care the method works, and it does not require extra transmitter power.

If the rise time of the pulse is also of importance, then the author can see no alternative to the use of extra transmitter power. The technique of Fig. 8(e) may be shown to be wasteful of power, as may the employment of a damping resistance in parallel with capacitor Y in Fig. 6. The most economical way of improving the rise time is to mis-match the probe and transmitter by increasing the value of capacitor X (Fig. 6). If the source resistance of the transmitter is R_s , then the effective resistance D placed across the tuned probe circuit by X is given by

$$D \approx 1/\omega^2 X^2 R_s. \quad (22)$$

On the other hand, the quiescent probe impedance, as viewed from the transmitter is

$$R_p \approx R_s D / Q \omega_0 L \approx R_s Q_{\text{eff}} / Q, \quad R_p \ll R_s, \quad (23)$$

where Q_{eff} is the effective quality factor of the loaded probe. Thus for power W to be dissipated in the probe, the emf generated by the transmitter must be

$$E \approx (WR_s Q / Q_{\text{eff}})^{1/2}, \quad (24)$$

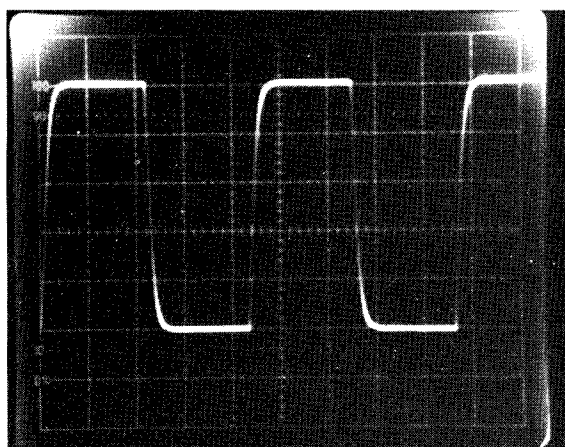
whereas if the probe were matched, the emf would be

$$F = 2(WR_s)^{1/2}. \quad (25)$$

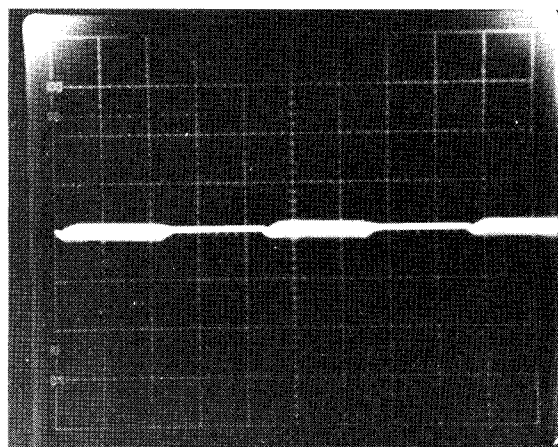
Thus to decrease the pulse rise (and fall) time by a factor σ as compared with the matched case ($Q_{\text{eff}} = Q/2$),

$$\frac{E^2}{F^2} \approx \frac{Q}{4Q_{\text{eff}}} = \frac{\sigma}{2}, \quad (26)$$

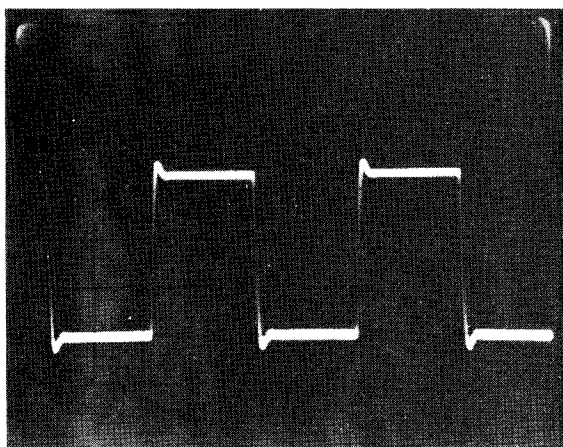
or in other words, the power specification of the transmitter must be increased by a factor $\sigma/2$. A similar calculation reveals that employment of a damping re-



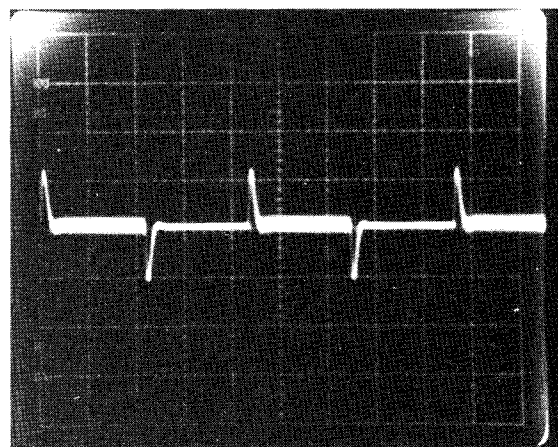
(a)



(b)



(c)



(d)

FIG. 9. Phase-glitch occurs when the probe is incorrectly tuned. The traces, which were obtained by phase-sensitive detection in quadrature of the voltage across the probe,¹¹ show quite clearly that no quadrature component is present when the probe is exactly on resonance (a and b). However, a considerable glitch exists off-resonance. Traces (c) and (d) were obtained at the -3 dB point. The transmitter was a broadband system; if the transmitter is tuned, glitch is very difficult to avoid. The width of the glitch is directly related to τ , the time constant of the probe (see Fig. 8).

sistance requires an increase in the power specification by a factor of 2σ . Mismatching the probe, as opposed to damping it, therefore saves a factor of four in transmitter power specification. The transmitter must, however, be stable when the output is effectively short-circuited. This is generally the case with commercial broadband amplifiers.

APPENDIX: DESIGN NOTES

As emphasis has been placed in the text on understanding, rather than finer design points, the following notes may be of assistance in building a suitable probe and preamplifier.

Preamplifier: A range of I_{DSS} from 13 to 30 mA was encountered with GAT 1 FETs. The drain resistances of the FETs may therefore need to be altered from the values shown. The 45° phase shift networks on the first and third stages have a broad second order response which makes them non-critical and returns the amplifier phase shift at high frequencies to -180° . This decreases the likelihood of oscillation at the roll-off frequency. (The 45° shifts can of course be reduced if device phase shifts are significant.) The variable feedback capacitor was two parallel pieces of 18 gauge wire each about 1 cm long. The amplifier was built on single-sided p.c. board using rf layout techniques. The shield, which was essential, bisected the bottom of the first FET touching the FET case. The output stage was broadly tuned on the assumption that an intermediate frequency is used in the receiver. The tuning rejects unwanted sideband noise. If desired, a broadband transformer may be substituted. The output impedance is nominally 50 Ω ; it may be changed by altering the emitter resistor of the output transistor.

Attenuator: Various attenuator designs were explored, including the use of JFETs, enhancement mode FETs, CMOS analogue switches and PIN diodes. All in some manner degraded the noise performance, uhf JFETs being the least offensive. At the low frequencies and high impedances used, ordinary fast-recovery signal diodes were found to be best in the gate configuration shown. It is vital that the leads of the protection diodes be

kept as short as possible and that attention be paid to the layout of the attenuator to optimize ground return paths. Unwanted inductance can cause the preamplifier input to be overloaded.

Switching Logic: The rectified input from across the protection diodes operates into a Schmitt trigger. To ensure that rise and fall times are fast resistor A (Fig. 6) may need adjustment. The delays in the positive and negative sections of the switching must be equal so that the outputs to the attenuator are released to float down simultaneously. Delay may be adjusted by changing resistor B. Float-down time is adjusted with capacitor C and symmetry of float-down with resistor D. Substantial errors in delay and float-down symmetry may result in pre-amplifier overload.

Probe: Several sets of crossed diodes in parallel may be needed to handle the current, depending on the transmitter power and diodes used. When setting up, the receive mode is tuned first by monitoring for maximum signal and/or minimum noise output from the preamplifier, a very small signal being injected, for example, via the capacitance of the crossed diodes on the transmitter side. The transmit mode is tuned with capacitor Y by monitoring the voltage across the probe coil by placing an oscilloscope probe near the coil. When applying an inverted final pulse to speed ring down, the presence of a null (see Fig. 8) in the ringing pattern is an excellent guide to correct tuning.

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